

MATH Unit 1 Previous Year Questions

Mathematics (MATH) · Unit 1 · Matrices & Linear Algebra · 2025

Questions collected from previous year university examinations (2021–2025).

Part A — Short answer (2 marks)

1. Compare Rank of a Matrix with Diagonalization. When would you prefer one over the other? [2023]
2. Write short notes on Systems of Linear Equations. [2025]
3. Explain Eigenvalues and Eigenvectors with a neat diagram and a suitable example. [2024]
4. Compare Cayley-Hamilton Theorem with Rank of a Matrix. When would you prefer one over the other? [2025]

Part B — Long answer (10 marks)

1. Define Diagonalization. Derive or justify the key properties involved. [2021]
2. Define Quadratic Forms. Derive or justify the key properties involved. [2024]
3. Explain Rank of a Matrix with a neat diagram and a suitable example. [2023]
4. Illustrate Systems of Linear Equations with a worked example. [2021]
5. What is meant by Eigenvalues and Eigenvectors? Discuss its advantages and limitations. [2023]
6. Explain Cayley-Hamilton Theorem with a neat diagram and a suitable example. [2023]
7. What is meant by Diagonalization? Discuss its advantages and limitations. [2024]
8. Discuss the role of Quadratic Forms in Matrices & Linear Algebra. [2023]
9. What is meant by Rank of a Matrix? Discuss its advantages and limitations. [2022]
10. What is meant by Systems of Linear Equations? Discuss its advantages and limitations. [2021]